BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming Challenge and Review

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. A GBP 5-per-ticket program uses total = tickets + 5. What is the correction?

2. Insert 6 into sorted [2, 5, 8]. What is the result?

A) [2, 5, 8, 6]
B) [2, 5, 6, 8]
C) [6, 2, 5, 8]
D) [8, 6, 5, 2]

3. "A comparison operator must always produce a text paragraph." Correct this. Explain why.

4. A warning needs high temperature OR high pressure. "OR is false when both inputs are true." Correct this. What happens when both are high?

Today's objectives

Independently design and develop a small program. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Planning
Define inputs, processing, output and success criteria.

Selection
Use IF or ELSE to choose valid outcomes.

Iteration
Repeat actions with a bounded FOR or condition-controlled WHILE.

Testing
Check normal, boundary and invalid inputs, then refine.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Planning	Define inputs, processing, output and success criteria. For example, consider a club booking tool.
Selection	Use IF or ELSE to choose valid outcomes. For example, consider a revision calculator.
Iteration	Repeat actions with a bounded FOR or condition-controlled WHILE. For example, consider a rewards app.
Testing	Check normal, boundary and invalid inputs, then refine. For example, consider a canteen voucher.

I Do - One worked example

A club booking tool is being developed. Explain how planning and selection are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Planning [1]; Define inputs, processing, output and success criteria. [1].

3. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

4. Give one accurate point and a linked reason for each technique.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A revision calculator is being programmed. Explain why planning matters. [2 marks]

2. For a rewards app, explain how selection is used. [2 marks]

3. A canteen voucher needs iteration and testing. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing condition to report a hot sensor.

```
sensor = 29
if ... then
print("Too hot")
else
print("OK")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection and iteration can help a fundraising tickets. [4 marks]

Q4. Explain the roles of planning, iteration and testing when creating a quiz score. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A useful algorithm plan identifies inputs, processing, outputs and success criteria. ☐ T ☐ F
2. Define inputs, processing, output and success criteria ☐ T ☐ F
3. Use IF or ELSE to choose valid outcomes ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Use total = tickets * 5; addition does not calculate GBP 5 for each ticket. ☐

Q2 B - [2, 5, 6, 8] ☐

Q3 A Boolean result: true or false. ☐

Q4 OR is true if at least one input is true, including both. AND needs both true. NOT reverses a value. ☐

Code First - Model

```
sensor = 29
if sensor > 28 then
print("Too hot")
else
print("OK")
endif
```

Output: Too hot

We Do - Check the model

1. Planning [1]; Define inputs, processing, output and success criteria. [1]. ☐

2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1]. ☐

3. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Planning [1]; Iteration [1]. ☐

Q3 Selection [1]; Use IF or ELSE to choose valid outcomes. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. ☐

Q4 Planning [1]; Define inputs, processing, output and success criteria. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: